DAYCLI: Production and reporting of daily summaries for surface fixed land stations

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Executive Summary

These guidelines support National Meteorological and Hydrological Services (NMHSs) in the production and international exchange of daily climate data using the DAYCLI BUFR message. DAYCLI is designed as a climatological data exchange product. It is intended to provide the temporal granularity needed to share information on extremes, which are of primary importance in monitoring climate variability and change, and as such complements real-time observational messages and monthly climate summaries, and aligned with official national climate records.

In operational practice, NMHSs define and distribute daily climate values using local, regional or national Attribution day, referenced to a Local Meteorological Time Zone (LMTZ). Preservation of this reference time is essential to maintain the homogeneity and continuity of long-term daily climate series. It also ensures consistency between internationally exchanged data and national climate datasets.

Experience with the original DAYCLI BUFR sequences **3-07-074 and 3-07-075**, as defined in the *Manual on Codes (WMO-No. 306)*, highlighted limitations related to UTC-based time referencing and insufficient alignment with national climatological practices. These limitations can introduce ambiguity in the interpretation of Attribution days. They can also increase processing complexity for NMHSs, and create risks of inconsistency in long-term climate series.

To address these limitations, an amended DAYCLI BUFR **sequence** (**3-07-095**) **was developed**. This revised approach improves the alignment of DAYCLI with NMHS climatological workflows and practices. It also provides a clearer framework for the international exchange of nationally validated daily climate datasets than the previous sequences did.

The guidelines are centered on Climatological Data Management System (CDMS) workflows used by NMHSs and support progressive implementation. Differences in national capacities are explicitly recognized, and the absence of some metadata or quality information should not prevent participation in international exchange.

Quality control remains a national responsibility as do the quality control (QC) flags used by NMHSs in their CDMSs. To support interoperability and transparency, the DAYCLI message includes standardized DAYCLI QC flags, which are independent of the flags used by NMHSs. The usage of the DAYCLI quality control flags inside the DAYCLI message is strongly encouraged, as they highlight situations where extremes, or ambiguous situations such as rainfall aggregations, are present. Nevertheless, the absence of DAYCLI QC flags inside the DAYCLI message does not imply the absence of national quality control, and should not prevent international exchange.

Together, the amended BUFR sequence and the associated technical guidelines provide a robust and practical framework for the global exchange of daily climate data. They improve clarity, interoperability and interpretability at regional and global levels, while fully respecting national climatological practices and responsibilities.

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1 Purpose of the DAYCLI Guidelines

The purpose of these guidelines is to support National Meteorological and Hydrological Services (NMHSs) in implementing the DAYCLI BUFR message for the international exchange of daily climate data derived from surface land stations.

The guidelines are intended primarily for **NMHS climatology departments** and focus on climatological and operational aspects of DAYCLI implementation rather than on detailed BUFR encoding instructions. Their objective is to translate the DAYCLI technical specification into practical, NMHS-oriented guidance. This guidance can be applied within existing national operational environments.

DAYCLI is designed to support the exchange of **nationally validated daily climate data** produced under the responsibility of NMHSs. It is not intended for near-real-time observational reporting, but rather to complement existing data exchange mechanisms by addressing the specific needs of climate monitoring, assessment and services.

In a variable and changing climate, extreme events often exert a disproportionate impact on climate-sensitive activities and sectors. The monitoring of daily climate extremes, such as temperature maxima and minima and intense precipitation, is therefore increasingly important for climate monitoring, assessment and services, and represents a key motivation for the international exchange of daily climate data.

These guidelines complement existing WMO regulatory and technical material, including the Manual on Codes (WMO-No. 306, Volume I.2) and the Guide to Instruments and Methods of Observation (WMO-No. 8, Volume I). They do not introduce new regulatory requirements, and do not prescribe national observing, processing or quality control practices.

Key terms used throughout these guidelines are defined in a glossary provided in Annex A.

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2 Target audience and scope

These guidelines are primarily intended for **NMHS climatology departments** responsible for the production, validation, management and archiving of daily climate data derived from surface land stations.

Secondary users may include:

- staff responsible for metadata management and observing systems;
- information technology and data exchange specialists involved in BUFR implementation;
- regional and global climate data centres collecting and redistributing daily climate data;
- users of internationally exchanged daily climate datasets for climate monitoring, assessment and services.

The scope of these guidelines is limited to **daily climate data from surface land stations** and to the use of the DAYCLI BUFR message for international exchange. The document does not address marine, upper-air or remote-sensing observations.

These guidelines do not prescribe national observing practices, data processing methods or quality control procedures. They recognize that NMHSs operate under diverse technical, institutional and regulatory environments, and accommodate existing practices for the generation and exchange of daily climate parameters. They aim to support interoperability while preserving national responsibility for climatological practices.

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3 Conceptual overview of DAYCLI

In most NMHSs, daily climate data are derived from observations collected at surface land stations and processed within a **Climatological Data Management System (CDMS)**. The CDMS constitutes the authoritative system for the production, validation and archiving of the national climate record.

A DAYCLI message is a set of daily observations from surface land stations, submitted monthly, along with supporting metadata.

DAYCLI is designed to support the **international exchange of nationally validated daily climate datasets**, rather than the exchange of raw or near-real-time observations. It enables NMHSs to disseminate **official daily climate values that are consistent with national records and climatological practices**. The datasets exchanged through DAYCLI are derived from observations processed within the CDMS. They are associated with clearly defined Attribution days and corresponding measurement time periods, in accordance with national climatological practices.

Within the WMO data exchange landscape, DAYCLI complements existing mechanisms such as real-time observational messages and monthly climate summaries (CLIMAT). It fills a specific gap by providing a standardised framework for the exchange of daily climate data. This framework includes the metadata and quality information required for correct climatological interpretation. The DAYCLI message is submitted monthly, to allow time for quality checking of the data.

DAYCLI supports climate monitoring, climate services, regional and global climate data aggregation, and long-term climate analyses and research. The design of the message reflects the needs of climatological applications, where data consistency, traceability and interpretability are essential, while fully preserving national responsibility for climatological definitions and practices.

DAYCLI prioritises interoperability and interpretability over data compactness or system-specific optimisation. DAYCLI does not replace national data management systems; it relies on them.

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4 Daily climate variables reported in DAYCLI

DAYCLI supports the international exchange of a core set of **daily climate variables** commonly produced, validated and archived by NMHS climatology departments as part of their routine climatological operations.

The variables currently supported include daily maximum air temperature, daily minimum air temperature, daily mean air temperature, daily precipitation totals, depth of fresh snow, and depth of total snow on the ground. These variables correspond to key indicators used in climate monitoring, assessment and services.

Daily climate values reported in DAYCLI shall reflect **national climatological practices**, including the definition of the **Attribution day** and the **Climatological day**, for each parameter.

The **Attribution day** is the calendar day to which an NMHS assigns daily climate values. The Attribution day is defined according to national climatological practices and referenced to a Local Meteorological Time Zone (LMTZ). An Attribution day is defined **for a given observing station and element**, recognizing that a national territory may include more than one LMTZ. The Attribution day provides the temporal reference used for daily climate data exchange and archiving.

For examples, the Attribution day for the Melbourne (Australia) meteorological station is defined as running from 00:00 LMTZ to 23:59 LMTZ, with LMTZ being 11 hours ahead of UTC. Similarly, the Attribution day for Paris-Montsouris (France) meteorological station is defined as running from 00:00 LMTZ to 23:59 LMTZ, with LMTZ equal to Universal Time Coordinated (UTC).

The **Climatological day** is the 24 hour period (for instance midnight to midnight, 6am to 6am, 8pm to 8pm etc.; local time) over which a daily parameter has been accumulated or measured. Examples include the minimum, maximum and mean temperature, precipitation accumulation etc. For a given station, the Climatological day may vary with parameter.

The distinction between Attribution day and Climatological day is fundamental for the correct interpretation of daily climate data.

Except for the measurement of the total snow depth, all other daily measurements are taken over a 24-hour period and are either accumulated (daily precipitation and daily fresh snow depth), averaged (daily mean temperature) or represent the daily extreme (daily maximum and minimum temperature). Total snow depth is not an accumulated measurement, but rather a measurement taken at a specific time of day.

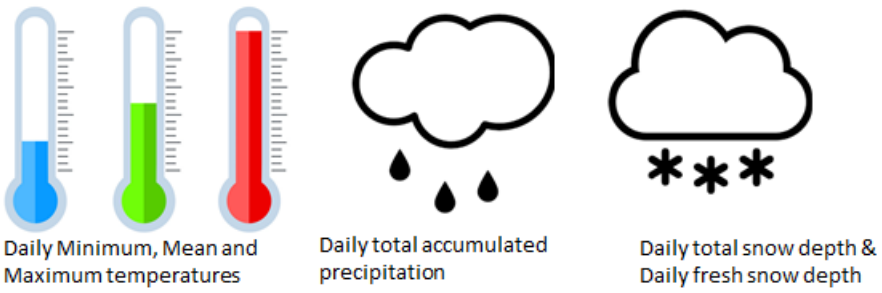


Figure-1: the 6 variables of the DAYCLI message

The 24-hour time period used to compute a parameter does not necessarily correspond to the Attribution day. For example, daily precipitation at the Paris-Montsouris station is not accumulated over the Attribution day (00:00–23:59 LMTZ), but over the period from 06:00 LMTZ of the Attribution day to 06:00 LMTZ of the following day (Attribution day +1).

Further explanations on the Attribution day and the Climatological day are provided in Sections 7 and 8.

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The reporting of daily climate variables in DAYCLI is flexible and does not require a complete set of parameters or metadata. Messages may therefore contain partial information.

Daily maximum, minimum and mean air temperature, precipitation and snow variables should be reported according to national data availability and practices. The information describing the **method used to compute daily mean temperature** is optional and should be provided where available to support correct climatological interpretation.

The absence of one or more daily climate variables for a given station or day does not prevent the exchange of a DAYCLI message. Missing values may reflect national observing practices, data availability or quality control outcomes.

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5 Structure of the DAYCLI message

The DAYCLI BUFR message is structured to support the exchange of daily climate data and does not reflect the structure of national systems such as CDMS databases. Its structure is not optimised for database storage and includes repeated information. This approach ensures that each message can be independently interpreted in an international context. It prioritises interoperability over storage efficiency.

Each DAYCLI message combines **essential metadata**, **daily climate values and associated quality information** within a single coherent structure. See the Figure-2 below. This ensures that exchanged data can be correctly interpreted and traced. It also reduces the need for additional external information.

The message structure is designed to preserve the context in which daily climate values are produced, including station identification, observational conditions and the temporal definition of the Climatological day and Attribution day. This is particularly important for the interpretation of daily extrema and accumulated quantities.

DAYCLI relies on BUFR sequences defined in the *Manual on Codes (WMO-No. 306)*. The structure has been designed to support climatological consistency, interoperability and long-term usability, while allowing NMHSs to implement the message within existing CDMS-centered workflows.

The Figure-2 shows the structure of the DAYCLI message.

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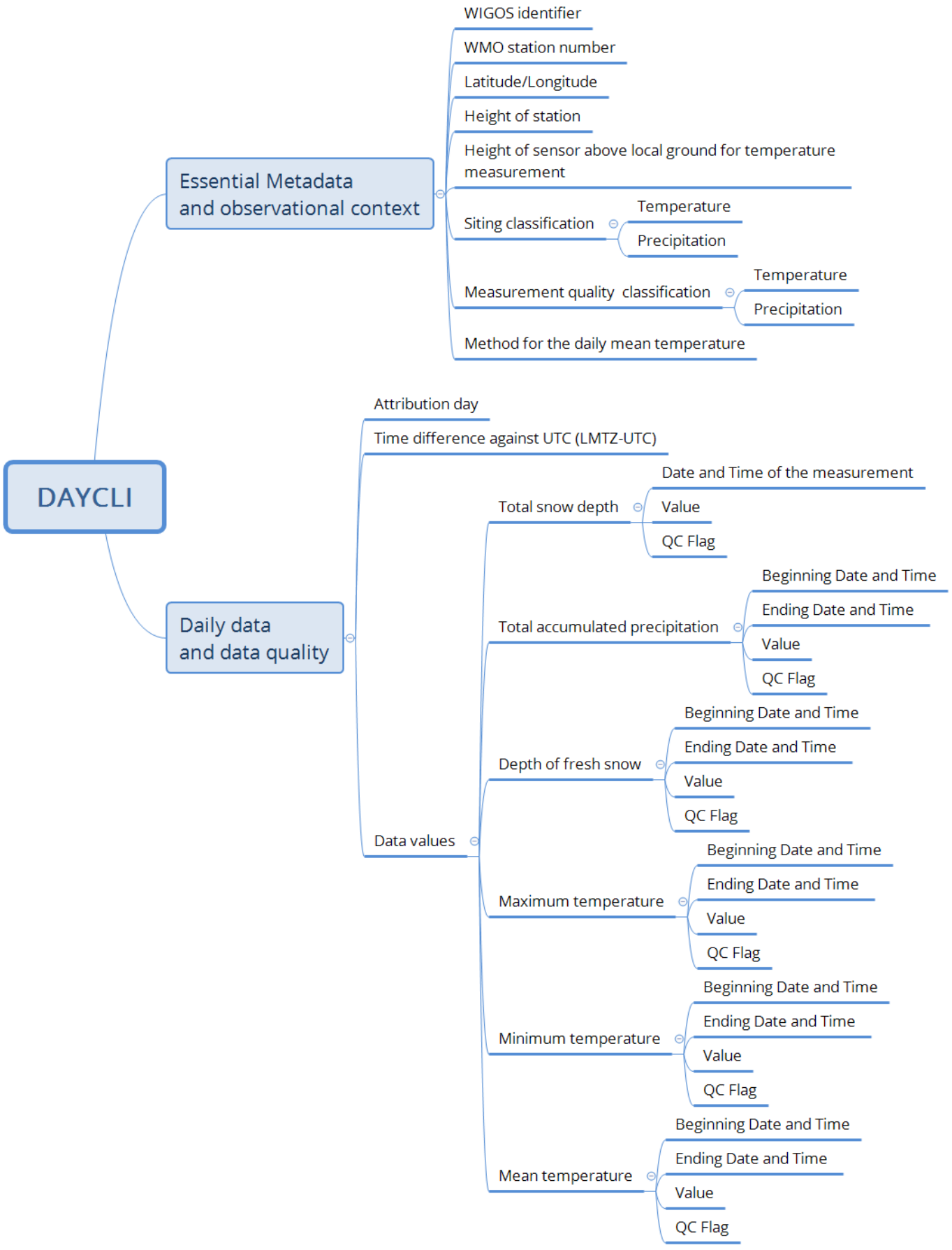


Figure-2: Structure of the DAYCLI message

6 Essential metadata and observational context

Daily climate values require appropriate metadata to describe the observational and environmental context in which they were produced. DAYCLI therefore supports the exchange of essential metadata together with the daily climate values to ensure transparency, traceability and climatological relevance.

Core metadata include a valid **station identifier**, **geographic location**, and information describing the observational context of the measurements. In particular, metadata related to temperature and precipitation observations provide important context for the representativeness of daily values, especially when data are compared across regions or used for long-term climate analyses.

For temperature observations, relevant metadata include the **height of the station**, the **height of the temperature sensor above ground level**, as well as the **siting classification** of the observing station. Siting classification is defined in the `Guide to Instruments and Methods of Observation (WMO-No. 8, Annex 1.D.)` and comprises five classes (Class 1 to Class 5), where Class 1 represents optimal siting conditions. This information helps users assess potential environmental influences affecting temperature measurements.

For both temperature and precipitation observations, DAYCLI supports the inclusion of a **measurement quality classification**, also defined in *WMO-No. 8 (Annex 1.G.)*. Measurement quality classification comprises four classes (A to D), where Class A represents the highest measurement quality. This classification reflects the overall quality of the measurement process and instrumentation and supports informed climatological interpretation.

The inclusion of siting classification and measurement quality classification is **strongly recommended** where available. However, their absence shall not prevent the exchange of DAYCLI messages. These metadata are intended to provide additional context for users rather than to impose constraints on data exchange, and in any case, sub-optimal observational conditions may still provide useful information for certain applications.

The following sub-sections describe in more detail the main metadata elements required within DAYCLI. Corresponding BUFR codes are provided for reference purposes only using the following Table-1 skeleton which is similar to tables used in WMO-No. 306:

BUFR Table Reference F X Y	Element Name	Element Description or/and value

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Table-1: table skeleton to represent BUFR element

With:

BUFR table Reference F X Y : data descriptors
Element Name: BUFR Name of the data
Element Description or/and value: Description and/or value of the BUFR data

6.1 WIGOS identifier

WIGOS Station Identifier (WSI) is a unique identifier used to register an observing station or platform in the OSCAR/Surface database. It replaces the legacy WMO identifiers (5-digit numbers for synoptic and climate stations), which had limited capacity. The WSI is structured into four blocks, allowing for an almost unlimited number of stations or platforms to be registered.

The WSI format is as follows: `[WIGOS Identifier Series]-[Issuer of Identifier]-[Issue Number]-[Local Identifier]`

Example: `0-20000-0-87585` for Buenos Aires Observatorio (Argentina)

The WSI is mandatory for WMO Members and is a critical component of the WMO Integrated Global Observing System (WIGOS). Detailed allocation rules and structure are provided in the `Manual on the WMO Integrated Global Observing System (WMO-No. 1160, Edition 2024, Attachment 2.2)` and the `Guide to the WMO Integrated Global Observing System (WMO-No. 1165, ~~Edition 2024~~, Chapter 9)`.

For GBON station (`WMO-No. 306, Volume I.2, GBON 1.2.1.2`), the sequence 3 01 150 shall not be set to missing and shall have the values corresponding with the station record in https://oscar.wmo.int/surface. Table-2 provides an example of WIGOS identifier.

BUFR Table Reference F X Y	Element Name	Element Description or/and value
3 01 150	WIGOS identifier	e.g. `0-20000-0-87585` for Buenos Aires Observatorio (Argentina)

Table-2: WSI for Buenos Aires Observatorio

6.2 WMO block and station number

The **"WMO block and station number"** refers to the traditional **5-digit numerical identifier** assigned by WMO to meteorological observing stations worldwide. This system was used before the introduction of the **WIGOS Station Identifier (WSI)**.

Definition:

WMO Block Number: The first two digits of the 5-digit number represent the **WMO regional block** assigned to a country or region. Each WMO Member is allocated one or more blocks of numbers for its stations.

Station Number: The last three digits identify the **individual station** within that country or region.

For GBON station (`WMO-No. 306, Volume I.2, GBON 1.2.1.3`), the sequence <3 01 001> shall report the traditional station identifiers (TSI) when available to ensure the continuity of data use, otherwise set to missing value. Table-3 provides an example of the WMO block and station number.

BUFR Table Reference F X Y	Element Name	Element Description or/and value

3 01 001	WMO block and station number	e.g. 96745 for Jakarta Observatory (Indonesia)

Table-3: WMO block and station number for Jakarta Observatory

6.2.1 Latitude/longitude (high accuracy)

In the DAYCLI BUFR, latitude and longitude have a standard precision of **0.00001 degrees** (approximately 10 meters at the equator).

The value in degree is **multiplied by 100 000** before being encoded as a signed integer to maintain precision.

The latitudinal and longitudinal positions of a station referred to in the World Geodetic System 1984 (WGS-84) Earth Geodetic Model 1996 (EGM96) (see the Guide to Instruments and Methods of Observation (WMO-No. 8, Volume I, Chapter 1, 1.3.3.2).

Example: Dakar Station YOFF (Senegal) (WSI= 0-20000-0-61641)

- **Actual Coordinates**:
  - Latitude: 14.73000°N
  - Longitude: -17.50000°E
- **BUFR Encoding**:
  - Latitude (code 0 05 001): 14.73330 × 100 000 = **1 473 000**
  - Longitude (code 0 06 001): -17.48330 × 100 000 = **-1 750 000**

BUFR code 3 01 021 is in fact the combination of the BUFR code 0 05 001 for the latitude and the BUFR code 0 06 001 for the longitude. Table-4 provides an example of latitude and longitude.

BUFR Table Reference F X Y	Element Name	Element Description or/and value
3 01 021	Latitude /longitude (high accuracy)	
0 05 001	Latitude (high accuracy)	1473330 # Latitude of Dakar (14.73000°N)
0 06 001	Longitude (high accuracy)	-1748330 # Longitude of Dakar (-17.50000°E)

Table-4: Latitude and longitude of Jakarta Observatory

6.3 Height of station ground above mean sea level

The Height of station ground above mean sea level refers to the vertical elevation of the ground at the location of a weather observing station, measured in meters above the mean sea level (MSL). This value is used to standardize observations and ensure accurate comparison of atmospheric data (such as pressure, temperature, and humidity) across different stations worldwide. The height is usually reported to the nearest whole meter (i.e., integer value).

References in the Guide to Meteorological Instruments and Methods of Observation, WMO-No. 8, Chapter 1. *Table-5 provides an example of the height of the station above mean sea level.*

BUFR Table Reference F X Y	Element Name	Element Description or/and value
0 07 030	Height of station ground above mean sea level (m)	2334 # Altitude of Thimphu (2 334 m)

Table-5: Height of station of Thimphu (Bhutan)

6.4 Siting and measurement quality classifications for temperature and precipitation

For the operational international exchange of daily climate data through the DAYCLI BUFR message, the inclusion of standardized metadata is essential to ensure that temperature and precipitation observations are interpretable, comparable and fit for use. Two key metadata elements are therefore recommended: **siting classification** and **measurement quality classification**.

Each precipitation sensor is associated with a siting classification and a measurement quality classification. Each sensor that measures temperature measurements is also associated with a siting classification and a measurement quality classification.

The siting classification provides information on the exposure and representativeness of the observing environment in which temperature and precipitation measurements are made. Its objective is to inform users about the potential influence of local conditions (e.g. obstacles, surface characteristics, slope, urbanization, shielding, gauge exposure) on the measured values. This classification supports the correct interpretation of daily extremes and totals, helps identify possible biases, and improves the usability of the data for climate monitoring, modelling, other climate services provision, and international exchange.

According to the Guide to Instruments and Methods of Observation (WMO-No. 8) (Annex 1.D), siting classification for surface observing stations on land is expressed using **five classes (Class 1 to Class 5)**:

- **Class 1** represents optimal siting conditions and the highest level of representativeness;
- **Classes 2 and 3** indicate progressively increasing potentially adverse influence from local environmental factors;
- **Classes 4 and 5** indicate significant siting limitations and lower representativeness.

The measurement quality classification describes the overall quality level assigned to the reported daily values based on applied observing practices, instrumentation, maintenance, and quality control procedures. Its objective is to indicate the confidence that can be placed in the measurement and to what extent it meets recommended WMO practices. This information enables users to assess data suitability for different applications, including climatological and climate change analyses, services, and international data exchange.

According to the Guide to Instruments and Methods of Observation (WMO-No. 8) (Annex 1.G), measurement quality is expressed using **four classes (A to D)**:

- **Class A** represents the highest measurement quality and full compliance with recommended practices;
- **Classes B and C** reflect decreasing levels of compliance or increasing uncertainty;
- **Class D** indicates the lowest level of measurement quality.

For DAYCLI implementation, these metadata elements are required for:

- **Air temperature** (maximum, minimum, and mean daily values), and
- **Precipitation** (daily totals).

Members should ensure that information on siting and measurement quality classifications for a sensor and a station are kept up to date.

Although provision of a classification for siting or measurement quality is strongly encouraged, their absence does not prevent the exchange of the DAYCLI message.

Detailed definitions, classification schemes, and implementation guidance shall follow the recommendations provided in the WMO reference material, in particular:

- Guide to Instruments and Methods of Observation (WMO-No. 8), including Annex 1.D and Annex 1.G
- Relevant provisions in the *Manual on Codes (WMO-No. 306)* and associated BUFR specifications.

Example for Temperature and Precipitation sensors <u>in a meteorological station</u>:

- **Temperature sensor**
  - Siting classification: Class 1
  - Measurement quality classification: Class B
- **Precipitation sensor (e.g. raingauge)**
  - Siting classification: Class 2
  - Measurement quality classification: Class C

Note: For the BUFR message siting and measurement quality classifications classes are combined and are assigned to a unique code. These are summarized in Annexes D and E. Table-6 provides examples of the siting and measurement classification for temperature and precipitation.

BUFR Table Reference F X Y	Element Name	Element Description or/and value
0 08 095	Siting and measurement quality classification for temperature	2 # Siting and measurement for temperature, Classes 1 and B, 1B equivalent to code **2**
0 08 096	Siting and measurement quality classification for precipitation	8 # Siting and measurement for precipitation, Classes 2 and C, 2C equivalent to code **8**

Table-6: Siting and measurement classifications

6.5 Method used to calculate the daily mean temperature

The DAYCLI message includes the information about the methodology used for computing the daily mean temperature at the station

In the descriptions below, the first sub-section describes the different methodologies to calculate the daily mean temperature typically employed by NMHSs. In the second sub-section, for information purposes, some known rules on allowable missing data used by NMHSs when calculating the daily mean temperature are listed.

6.5.1 Method types

This metadata gives the methodologies for computing mean temperature for DAYCLI messages.

The different methods, which are, so far, known to be used by WMO Members, are listed in Table-7. All methods use the available data to approximate the "true" daily mean (defined using the integral of the daily temperature curve); for example, the weighted means (codes 3 and 4) are attempts to estimate the full daily mean from a limited number of fixed-hour observations.

Code figure	Description
0	Average of maximum and minimum values: Tm = (Tx + Tn)/2 (see Note)
1	Average of the 8 standard synoptic observations taken every three hours
2	Average of 24-hourly values spaced one hour apart
3	Weighted average of 3 observations: Tm = (aT1 + bT2 + cT3) (see Note)
4	Weighted average of 3 observations and also maximum and minimum values: Tm = (aT1 + bT2 + cT3 + dTx + eTn) (see Note)
5	Automatic weather station complete integration from one-minute data (1440 minutes)
6	Average of the 4 observations taken six hours apart, and corresponding to intermediate synoptic times
7-254	Reserved
255	Missing value

Table-7: Code table 0 08 094 for calculation of daily mean temperature

Note: The letters a, b, c, d and e generically represent the weight associated with the respective temperature T. Tx and Tn represent respectively the daily maximum and minimum temperature. The sub-indices of T: 1, 2, 3 represent the values measured at different times 1,2 or 3.

The average should be calculated using the arithmetic mean of the indicated observations unless otherwise specified (i.e. code figures 3 and 4).

It is difficult to dictate a strict, general rule for all the stations worldwide, or even within a single country. Each Member's national meteorological and hydrological service (NMHS) must make their own decisions on this matter. Nevertheless, it is preferred that countries maintain a consistent calculation method over time. For example, a NMHS could decide to continue to exchange temperature data from a station that has been calculating mean temperature for over fifty years, using the 8 standard times for surface observations corresponding to the station's Climatological day (e.g. at 00, 03, 06, 09, 12, 15, 18 and 21 hour), thereby ensuring historical consistency of method.

Another NMHS could calculate the daily average temperature for some of its newly automated stations by integrating from one-minute data (i.e. the average of 1 440 values).

At the very least, the NMHS is advised to:

- ✓ continue to maintain homogeneous time series based on consistent calculation methods. Depending on the products or services required, a NMHS may maintain several time series for the same parameter and for the same station, but over different periods and with different methods. For example, an average daily temperature series calculated from 8 observations and another average time series calculated from 1440 observations. The NMHS is free to use either method in calculating mean temperature for DAYCLI.

- ✓ **always describe the calculation method used for data exchanged with DAYCLI**. Note that if the method used by an NMHS is not listed in the BUFR table 0-08-094 the NMHS could report it to WMO for addition.

Table-8 presents an example for **EL KHOMS (Libya), WSI =** 0-20000-0-62012, where the mean temperature is calculated from eight three-hourly observations.

BUFR Table Reference F X Y	Element Name	Element Description or/and value
0 08 094	Method used to calculate the average daily temperature	1 # Average of the 8 standard synoptic observations taken every three hours at EL KHOMS (Libya)

Table-8: Method used to calculate the average mean temperature

6.5.2 How to handle missing data

There is no uniform standard to handle missing data in the computation of a daily mean of a parameter. The rules are varied and depend on the calculation method used and local practices. The following are examples of rules used by NMHSs for managing missing data depending on the samples. These guidelines do not provide an exhaustive list of rules, but rather highlights their diversity. Note also that missing data may also impact on minimum and maximum temperatures and precipitation totals; this situation is dealt with in Section 11.9 below.

6.5.2.1 Daily mean using 1440 1-Minute values

There are various rules that are currently implemented in automatic weather stations (AWS), in data collection platforms (platforms that collect and manage Automatic Weather Station (AWS) data and Climate data management systems (CDMSs). One of them is the rule 1/10 explained below:

Rule 1/10

If 1/10 of the values or more are missing (i.e. 144 values or more out of the 1440 One-Minute values) then the corresponding daily data is considered to be missing.

6.5.2.2 Daily mean using 24 Hourly values

A variety of methods are used:

Rule 1/10

If 1/10 or more of the values are missing (i.e. 3 values or more out of the 24-Hourly values) then the daily mean shall be computed with the 8 values at synoptic standard times (e.g. at 00, 03, 06, 09, 12, 15, 18 and 21 hour). And if two (some NMHSs use only one) nor more values among the 8 values at synoptic standard times is missing the daily mean is considered as missing.

Some NMHSs are less restrictive and increase the percentage of missing values allowed to 1/6. Other allows to compute a daily mean with 4 values for main standard times (00, 06, 12,18) or 4 values from intermediate standard times (03, 09, 15, 21). While there is no provision for this in the DAYCLI BUFR code, it is strongly recommended that NMHSs document their methodologies in this regard so that users and researchers understand the methodology.

Rule 3 consecutive or 5 non-contiguous

If 3 consecutive values or more are missing among the 24-Hourly values then the daily mean shall be computed via the “Rule 2/8” below, using the 8-Hourly values at standard times.

Also, If 5 **global** values or more are missing among the 24-Hourly values then the daily mean shall be computed with the rule 2/8 with the 8-Hourly values at standard times (see the rule 2/8 below).

6.5.2.3 Daily mean using 8 3-Hourly values at standard time

Rule 2/8

if 2 or more Hourly values among the 8 values at standard times are missing then the daily mean is considered as missing

Rule 1/8

If 1 value among the 8 standard times for surface observations (00, 03, 06, 09, 12, 15, 18, 21) is missing then the mean shall be computed with the 4 main standard times (00, 06, 12, 18) or the 4 intermediate standard times (03, 09, 15, 21).

6.5.2.4 Daily mean using 4 6-Hourly values at main or intermediate standard time

Rule 1/4

If 1 among the 4 main standard times (00, 06, 12, 18) or if 1 among the 4 intermediate standard times (03, 09, 15, 21) is missing then the mean is considered as missing.

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7 Attribution day and difference between LMTZ and UTC

Within the DAYCLI message, NMHSs should transmit their data values in keeping with their climatological practices.

Many NMHSs organize their daily climatological observations, calculations and archives according to national or regional time standards and not necessarily on Coordinated Universal Time (UTC). The exchange of daily data through the DAYCLI message should be consistent with these official NMHS practices used in compiling climatological archives and daily datasets.

The **Attribution day** was briefly defined in Chapter 4 as the date to which an NMHS assigns daily climate values for the purposes of daily climate data exchange and archiving, in accordance with its local practices. More fully, it is the date (year-month-day) to which daily values of temperatures, precipitation, total snow depth and fresh snow depth are assigned for a specific station, expressed in terms of the "Local Meteorological Time Zone" (**LMTZ**).

In order to relate this LMTZ to UTC, the offset of the LMTZ against UTC shall be expressed in minutes for all variables.

It is important to recognize that the LMTZ applies to individual observing stations, and that many countries may include more than one LMTZ. This will affect the offset against UTC.

Example 1:

Table-9 shows the format for an observation of daily rainfall at Melbourne Airport Station. In Australia the Climatological day (see next Section) for 24 hour rainfall totals is defined as 9am to 9am, with the value attributed to the day on which this period ends. Also, the Local Meteorological Time Zone for Melbourne Airport station (WSI=0-20000-0-94866) is 11 hours (**660** minutes) in advance from UTC in the southern hemisphere summer.

BUFR Table Reference F X Y	Element Name	Value and / or Element Description
0 08 028	extended time significance	`40` #Set the next year-month-day to be the Attribution day in Local Meteorological Time Zone (LMTZ)
3 01 011	year-month-day	`20260115` #The Attribution day is here set to the 15 January 2026
0 08 025	time difference qualifier	`3` # Set the next time value to be the difference between LMTZ and UTC
0 26 003	(time difference in minutes)	`660` #Difference between UTC at Melbourne Airport is 660 minutes i.e. 11 hours

Table-9: Attribution day and time difference against UTC for Melbourne (Australia)

Draft

Example 2

- Table-10 similarly shows the format for the Attribution day for London Weather Centre station (WSI= 0-20000-0-03779) which is 0 hour (**0** minutes) in advance from UTC.

BUFR Table Reference F X Y	Element Name	Value and / or Element Description
0 08 028	extended time significance	`40` #Set the next year-month-day to be the Attribution day in Local Meteorological Time Zone (LMTZ)
3 01 011	year-month-day	`20260115` #The Attribution day is here set to the 15 January 2026
0 08 025	time difference qualifier	`3` # Set the next time value to be the difference between LMTZ and UTC
0 26 003	(time difference in minutes)	`0` #Difference between UTC at London Weather Centre is 0 minute i.e. 0 hours

Table-10: Attribution day and time difference against UTC for London (UK)

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8 Daily measurement time and measurement period (or Climatological day)

It is important to differentiate between a spot measurement taken at a specific time of day such as total snow depth and a 24-hour measurement period such as precipitation, fresh snow depth, mean temperature, maximum temperature, and minimum temperature.

8.1 Total snow depth: a specific time measurement

The Total snow depth does not represent an accumulation of snow during 24 hours, it is measured once a day at a specific time. Only the date and time is therefore necessary in the DAYCLI message.

Example

- Table-11 shows the format for the Total snow depth at Toulouse Blagnac station (WSI= 0-20000-0-03779) in France. This parameter is measured every day at 06:00 am LMTZ (which is equal to 06:00 am UTC).

BUFR Table Reference F X Y	Element Name	Value and / or Element Description
0 08 028	extended time significance	`42` #Set the next year-month-day to be the date of the measurement of the Total snow depth
3 01 011	year-month-day	`20260115` #Date when the measurement of the Total snow depth is made, here set to the 15 January 2026
3 01 012	hour-minute	`06:00` # Hour and minute when the measurement of the Total snow depth is made, here set to 06:00

Table-11: Climatological day definition for total snow in Toulouse

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8.2 Measurement period or Climatological day for the other variables

For all other variables, daily accumulated precipitation, depth of fresh snow, Maximum, Minimum and Mean temperature, the beginning date and time and the ending date and time that characterize the 24-hour period they measure must be specified. This defines the Climatological Day.

For some variables, such as temperature and precipitation, NMHSs often define a period of measurement (Climatological Day) that extends either before or after their Attribution day. As, historically, most stations were manual (and many national networks still contain a substantial proportion of manual observations), national observations were often influenced by which time is most practical for observers (e.g. observation times in the morning are much more common than midnight). The start date and time may occur on the day preceding the Attribution day for some variables; for others, the end date may occur in the day following the Attribution day.

For instance, a 24-hour accumulated precipitation attributed to day D may be based on a **Climatological day** starting at 09:01 am LMTZ on day D-1, and finishing at 09:00 am LMTZ on Day D. This was the case in the example shown in Chapter 7, and illustrated in Figure-3 below. So, in this case the 24 hour Climatological day is defined as 09:01 LMTZ day D-1 to 09:00 LMTZ day D, and the Attribution day is D (or 09:00 LMTZ day D-1 to 08:59 LMTZ on day D, if that is the prevailing practice at the NMHS).

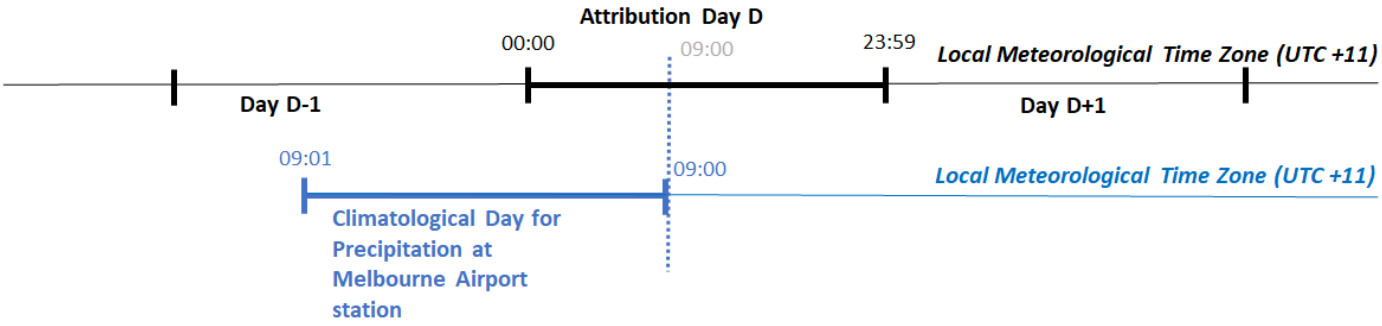


Figure-3 – Climatological day for precipitation at Melbourne Airport station (Australia). The 24 hour rainfall total measured at 09:00 on the Attribution day consists of the cumulative total over the period extending from 09:01 on Day D-1 to 09:00 on Day D.

Another example, when the Climatological day for a station extends after the Attribution day (D+1). That is the case for example for the Toulouse Blagnac station (France) for the Maximum Daily Temperature. The Climatological day here for the Maximum Temperature is defined as 06:01 LMTZ day D to 06:00 LMTZ day D+1 with the Attribution day being D as illustrated by the Figure-4 below:

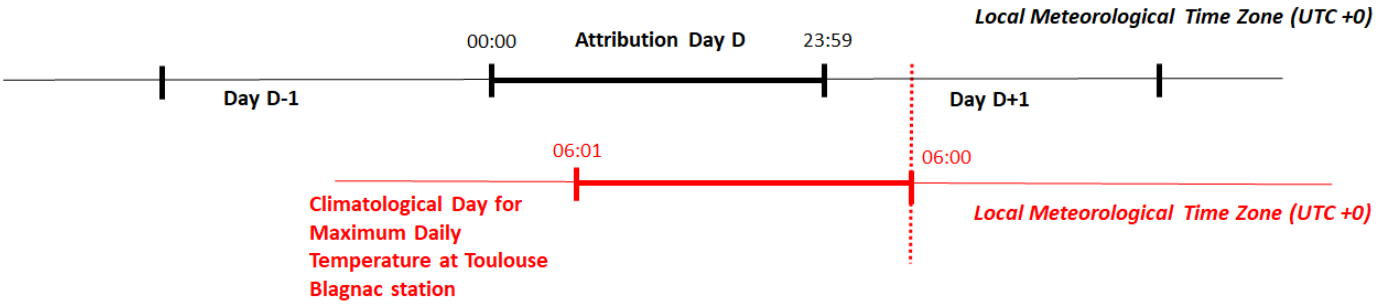


Figure-4 – Climatological Day for Maximum Temperature at Toulouse Blagnac (WSI = 0-20000-0-07630, France)

Table-12 presents an example, showing the format for all DAYCLI variables for the station of **SAINT PIERRE** (WSI= 0-20000-0-71805) for the 15 January 2026.

SAINT PIERRE is 4 hours before UTC.

- Total snow depth is observed every day at 08:00 LMTZ
- Accumulated precipitation is from Day D at 08:01 to Day D+1 at 08:00 LMTZ
- Depth of fresh snow is from Day D at 08:01 LMTZ to Day D+1 at 08:00 LMTZ
- Maximum temperature is from Day D at 08:01 LMTZ to Day D+1 at 08:00 LMTZ
- Minimum temperature is from Day D-1 at 08:01 LMTZ to Day D at 08:00 LMTZ
- Mean temperature is from Day D at 00:01 LMTZ to D D+1 at 00:00 LMTZ

BUFR Table Reference F X Y	Element Name	Value and / or Element Description
0 08 028	extended time significance	**40** #Set the next year-month-day to be the Attribution day in Local Meteorological Time Zone (LMTZ)
3 01 011	year-month-day	**20260115** #The Attribution day is here set to the 15 January 2026
0 08 025	time difference qualifier	**3** # Set the next time value to be the difference between LMTZ and UTC
0 26 003	(time difference in minutes)	**-240** #Difference between UTC at London Weather Centre is -240 minutes i.e. -4 hours
0 08 028	extended time significance	**42** #Set the next year-month-day to be the date of the measurement of the Total snow depth
Total snow depth		
3 01 011	year-month-day	**20260115** #Date when the measurement of the Total snow depth is made, here set to the 15 January 2026
3 01 012	hour-minute	**08:00** # Hour and minute when the measurement of the Total snow depth is made, here set to 08:00
Total accumulated precipitation		

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BUFR Table Reference F X Y	Element Name	Value and / or Element Description
0 08 028	extended time significance	**41** #Set the next dates to be the beginning date and the ending date of the measurement period (Climatological day)
3 01 011	year-month-day	**20260115** #Beginning Date of the period for the Total accumulated precipitation measurement
3 01 012	hour-minute	**08:01** # Beginning Time of the period for the Total accumulated precipitation measurement
3 01 011	year-month-day	**20260116** #Ending Date of the period for the Total accumulated precipitation measurement
3 01 012	hour-minute	**08:00** # Ending Time of the period for the Total accumulated precipitation measurement
	Depth of fresh snow	
3 01 011	year-month-day	**20260115** #Beginning Date of the period for the Depth of fresh snow measurement
3 01 012	hour-minute	**08:01** # Beginning Time of the period for the Depth of fresh snow measurement
3 01 011	year-month-day	**20260116** #Ending Date of the period for the Depth of fresh snow measurement
3 01 012	hour-minute	**08:00** # Ending Time of the period for the Depth of fresh snow measurement
	Maximum temperature	

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BUFR Table Reference F X Y	Element Name	Value and / or Element Description
3 01 011	year-month-day	**20260115** #Beginning Date of the period for the Depth of fresh snow measurement
3 01 012	hour-minute	**08:01** # Beginning Time of the period for the Depth of fresh snow measurement
3 01 011	year-month-day	**20260116** #Ending Date of the period for the Depth of fresh snow measurement
3 01 012	hour-minute	**08:00** # Ending Time of the period for the Depth of fresh snow measurement
	Minimum temperature	
3 01 011	year-month-day	**20260114** #Beginning Date of the period for the Depth of fresh snow measurement
3 01 012	hour-minute	**08:01** # Beginning Time of the period for the Depth of fresh snow measurement
3 01 011	year-month-day	**20260115** #Ending Date of the period for the Depth of fresh snow measurement
3 01 012	hour-minute	**08:00** # Ending Time of the period for the Depth of fresh snow measurement
	Mean temperature	
3 01 011	year-month-day	**20260115** #Beginning Date of the period for the Depth of fresh snow measurement
3 01 012	hour-minute	**00:01** # Beginning Time of the period for the Depth of fresh snow measurement
3 01 011	year-month-day	**20260116** #Ending Date of the period for the Depth of

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BUFR Table Reference F X Y	Element Name	Value and / or Element Description
		fresh snow measurement
3 01 012	hour-minute	00:00 # Ending Time of the period for the Depth of fresh snow measurement

Table-12: Concrete example of layout for all DAYCLI variables for Saint Pierre Station the 15/01/2026

9 Quality control and validation workflow

Quality control is a fundamental component of daily climate data production and remains a **national responsibility** under the authority of NMHS climatology departments. Quality control procedures are defined and applied according to national climatological practices and operational constraints at NMHS level.

In most NMHSs, daily climate data quality control is implemented within a **Climatological Data Management System (CDMS)**. Ideally the CDMS supports automated checks, expert review and, where applicable, consolidation of observations from multiple sources before daily values are validated and archived as official national climate datasets.

There is currently **no globally standardized climatological quality control flagging system** for daily climate data. National practices may differ in terms of quality control rules, thresholds, flags and expert intervention. These differences reflect climatic diversity, observing systems, historical practices, staffing availability, and institutional requirements across NMHSs.

DAYCLI does not prescribe quality control methodologies or thresholds.

Instead, it is designed to support the **transparent exchange of the outcomes of national quality control processes**, while preserving the meaning and authority of national assessments.

Quality control outcomes established within the CDMS may be expressed using national or system-specific flags and codes. Where feasible, these national flags may be **mapped to DAYCLI quality control flags** to support interoperability and common understanding at regional and global levels.

The absence of a one-to-one correspondence between national quality control flags and DAYCLI flags shall not prevent data exchange. NMHSs are encouraged to implement such mappings progressively, according to their technical capacity and operational priorities.

The quality control and quality assurance practices that may be applied to daily climate data are documented in existing WMO guidance material. In particular, NMHSs are encouraged to refer to the *WMO Guidelines on Surface Station Climate Data Quality Control and Quality Assurance for Climate Applications (WMO-No. 1269)*, the *Guide to Climatological Practices (WMO-No. 100)*, and the *Climate Data Management System Specifications (WMO-No. 1131)* for detailed methodological and system-level guidance.

These DAYCLI guidelines do not duplicate or replace such documents, but provide a framework for representing the outcomes of national quality control processes in international data exchange.

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10 Representation of quality information and QC flags

DAYCLI supports the representation of **quality information** associated with daily climate values in order to facilitate transparent and interoperable interpretation of the confidence placed in, and contingencies (such as known aggregations) applying to, exchanged data at regional and global levels.

Quality information in DAYCLI is expressed through **quality control (QC) flags** that describe the outcome of national quality control and validation processes applied within the CDMS. These flags provide users with an indication of the confidence that can be placed in a reported daily value and of any known limitations affecting its use.

QC flags in DAYCLI are designed to support interoperability. They do not replace national quality control practices or expert judgement; in fact they rely on these. They provide a common framework for communicating quality information while preserving the meaning and authority of national assessments.

National or CDMS-specific QC flags and codes may differ in structure, number and semantics. Where feasible, NMHSs may establish a **mapping** between their national QC flags and the DAYCLI QC flags to support consistent international interpretation. Such mappings, where implemented, are expected to be defined and maintained at national level.

The absence of a QC flag associated with a reported daily value does not imply that the value has not undergone quality control. It may reflect limitations in mapping national QC information to the DAYCLI representation or a progressive implementation approach. The absence of QC flags shall not prevent the exchange of DAYCLI messages.

QC flags should be interpreted in conjunction with other metadata and contextual information, including siting classification, measurement quality classification and national climatological practices. Users of DAYCLI data are encouraged to consider quality information as guidance rather than as an absolute measure of data usability.

The DAYCLI quality control flags are listed in the table13 below:

Code Quality Flag	Meaning
0	Data checked and declared good
1	Data checked and declared suspect
2	Data checked and declared aggregated
3	Data checked and declared out of instrument range
4	Data checked and declared aggregated and out of instrument range
5	Parameter is not measured at the station
6	Daily value not provided
7	Data unchecked
8…254	Reserved
255	Missing (QC info not available)

Table-13: QC flags of DAYCLI (BUFR Table 0 31 021 Associated field significance)

Note: Some NMHSs use the terminology “Suspect” to mean there is considerable doubt about the value, and it would therefore not be considered suitable for exchange, or even in local product development. However “Suspect” in the DAYCLI context indicates that while there may be some doubt that the value is an unqualifiedly accurate representation of the observation, confidence in the value is sufficient to permit exchange and use.

In this classification, “Data checked” indicates that the data have been checked by personnel or systems assigned the relevant authority, and an adjudication made as to whether the data are declared to be fit (or unfit) for at least immediate operational use. If QC is not carried out, then the “Data unchecked” flag should be assigned.

These QC flags are intended to support interpretation and transparency, not to define acceptance or rejection criteria for climate data use. The definitions of the QC flags are given below.

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10.1 Data checked and declared "Good"

Checks inspire confidence that the value assessed is a true representation of the variable for that day. A piece of data (datum) may fail one QC test, but confirmatory evidence may exist that the datum is still reliable. For instance, an extremely high temperature value is supported by similarly extreme values at neighboring stations.

10.2 Data checked and declared "Suspect"

Observation fails more than one QC test, is on the whole regarded as probably accurate, but there is some minor doubt. Overall, the operator would have at least medium confidence in the data, so it should still be exchanged.

Note: Where QC checks indicate low confidence in the value, or it is known to be wrong, the value should be regarded as Doubtful or Wrong. Such data should not be exchanged; instead it should be coded as "missing" (Code 6).

10.3 Data checked and declared "Aggregated"

In this case the amount of precipitation assigned for a day actually represents the amount of precipitation accumulated over several days. This situation is commonly associated with manual stations where, for instance, the gauge is not read over a weekend period. It can also occur for fresh snow amount, including where snow blocks a gauge.

It may also happen that a temperature value (minimum or maximum) may represent the lowest/highest value across a period of several days. In such cases, unless it is obvious which day the extreme occurred on, "Aggregated" is recommended.

10.4 Data checked and declared "Out of instrument range/instrument failure"

This covers the special case where the total amount of precipitation cannot be determined due to exceeding the capacity of an instrument such as a raingauge (or instrumental failure), but a lower bound is known, i.e. the capacity of the gauge, or the total at which instrument failure occurs. This situation typically applies in heavy rainfall situations which have obvious climatological and hydrological significance, and where the gauge is manually read. In such cases it is useful to record the fact that a significant precipitation event occurred, the exact amount of which is unknown, but at least exceeds a known threshold (i.e., the capacity of the gauge).

Such a flag could also be considered in cases where an automated instrumentation-related failure (e.g. loss of communications, etc.) enables a lower bound to be established; for instance, during a rainfall event where precipitation has occurred, but the instrument fails before the end of the rain event. In such cases it is at the discretion of the NMHS as to whether this approach should be used, or whether the value is simply flagged as "missing[i]". See the Use Case example 11.4 below.

Of course, it may also apply when an instrument is destroyed, by either the event itself or some other cause. In this case the "Out of instrument range/instrument failure" will likely be followed on subsequent days by "Daily value not provided" (Code 6 in Table-13).

10.5 Data checked and declared aggregated and out of instrument range

As the name implies, this is a combination of the previous two situations, flagging a situation where the amount of precipitation assigned for a day represents an accumulated total spanning two or more days, **AND** the situation represents an overflowed gauge /underestimated total case.

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10.6 Parameter is not measured at the station.

The station does not have the instrumentation to be able to measure this parameter. E.g. no instrument for measuring snow. This classification would also apply if the instrumentation is available, but is non-operational at that date due to maintenance, damage or some other cause.

10.7 Data value not provided.

The daily value is missing. There are several possible reasons for this: - transmission problem; observer absent; a doubtful value that the data provider considers so unlikely that it is not scientifically justifiable to provide it (see above under note "*Data checked and declared Suspect*"

10.8 Data unchecked

No QC processes have been performed on the data other than perhaps AWS Central Processing Unit tests, database ingest checks, etc. which will normally be confined to format or constraint tests, and thus will only catch gross errors, e.g., data that are physically impossible, or nearly so. When assigning "Data unchecked" to a value to be reported in DAYCLI messages, the implication is that no additional QC checking has been performed on the data except format or constraint tests, that is, there have been no checks on internal consistency, statistical (range) checks, temporal or spatial consistency checks, or other checks as described in WMO No 1269, Annex 1.

10.9 Quality control information is unavailable.

A data value is available, but no information is provided on quality control, even as to whether the value has been checked or not.

To assist Members, some examples of typical situations, and recommendations on how to assign Quality Flags, are provided in the next Section.

Note: Notwithstanding the requirement to provide DAYCLI reports within the standard period, to be defined, NMHSs are encouraged to provide updated quality assessments at a later stage. This is to ensure that the information available incorporates updated quality control or new information.

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11 Illustrative use cases flag assignment

This section provides **illustrative use cases** to support a common understanding of how quality control (QC) flags may be assigned and interpreted in typical climatological situations. The use cases are indicative and do not prescribe national quality control decisions, or the way they are flagged.

Use cases illustrate situations frequently encountered in daily climate data production, such as missing observations, partial data availability, aggregation of sub-daily observations into daily values, instrument limitations, snow-related effects on precipitation measurements, and extreme or unusual climate events.

In some cases, a daily climate value may be flagged to indicate known limitations while still retaining significant climatological value. For example, a daily precipitation total affected by snowfall or wind undercatch may be flagged to document the limitation without excluding the value from climate analyses.

The assignment of QC flags should reflect **national expert judgement** and established climatological practices. The use cases presented here are intended to support consistent interpretation at regional and global levels, rather than to define acceptance or rejection criteria for daily climate data.

Note: it is important to distinguish between situations where:

- a parameter (e.g. snowfall) is not measured;
- measurements are generally made, but a daily value is not provided; or
- QC information is not provided (i.e. QC missing = flag 255 in schema).

11.1 Use case: Parameter not measured at the station

Typical examples of such cases include, for instance, snow in a tropical lowland country, or snow is only measured seasonally. In that case, the QC code is 5: Parameter is not measured at the station.

Coding:

Depth snow value	**Associated QC information**	**QC code**
Missing	Parameter is not measured at the station	5

11.2 Use case: Parameter measured at the station, but the value for the day is missing

In this case, the daily value (in this case for maximum temperature) is missing and so of course QC cannot be carried out. Alternatively, QC was carried out, and the value is regarded as doubtful or wrong.

Coding:

Maximum Temperature	**Associated QC information**	**QC code**
Missing	Daily value not provided	6

11.3 Use case: QC not provided

Parameter is measured, daily value is present, but daily QC is missing. The QC code is 255. Note that this is separate from the case where there is an aggregation, as in Use case 5 (sub-Section 11.5).

It is also separate from the case where QC is not carried out (Code 7).

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Coding:

Maximum Temperature	Associated QC information	QC code
32.5	QC information not provided	255

11.4 Use case: Raingauge overflowed/out of instrument range

A daily value is measured, but there is an overflow for a raingauge, or perhaps a blocked gauge in the case of snow, and therefore the best we can say is that we can specify a minimum amount, but no more than that. In that case, the QC code is 3.

Coding:

Date	Precipitation	Associated QC information	QC code
Day 1	203 mm	Data checked and declared out of instrument range	3

Note that this classification might also be considered when obvious instrumental failure leads to a value or total that is almost certainly not representative of the daily total or extreme. For instance, a lightning-induced power failure stops recording of rainfall by an AWS, or some event stops measurements before the daily maximum/minimum temperature has been reached. The 2nd example below represents this situation.

Day 1	17.6 mm	Data checked and declared out of instrument range	3

Note:

In this case, the NMHS may determine that an instrumental failure has led to an unrepresentative value, but may still believe there is value in presenting the value. While such situations are likely to be rare, the NMHS may, at its discretion, regard the measurement as having some value, albeit heavily qualified. For instance, a rainfall value in an otherwise desert climate. In this case, it is clear that the value does not reflect the limits of the instrumentation, but rather that the instrument/sensor failed

11.5 Use case: Aggregation

In this case, observations are not available for some individual days, but the aggregated total (or highest/lowest value in the case of temperature) is known.

Example: In the case of precipitation, because of an observer problem or absence, the rain gauge (typically manually-read) has accumulated a precipitation amount over 4 days (from DAY 2 to DAY 5 in the example below). The precipitation amount for these 4 days is reported at DAY 5 (32.5 mm), following an earlier overflow situation. From DAY 2 to DAY 5 the QC flag i is set to 2 “Data checked and declared aggregated”, which allows an inference to be drawn on how many days the aggregation corresponds to. So, in this case, a total of 32.5mm has been measured over days 2-5, but it is not possible to determine the daily split. For Day 6, noting that the aggregation problem has been solved, this particular day reported no additional precipitation.

Note that if it is known for sure that no precipitation has occurred on one or more days during the aggregation period, it is recommended to include the value as 0.0, and the QC flag set to “Good” See also the example in Use Case 11.8 below. Alternatively, an NMHS may formulate a policy that partial disaggregations are not useful, and just define an aggregation for the entire period. As with all examples here, NMHS policies should define how the flags are allocated, but for transparency and useability, the method used should be consistent, and documented.

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Coding:

Date	Precipitation	Associated QC information	QC code
Day 1	203 mm	Data checked, and declared out of instrument range	3
Day 2	Missing	Data checked and declared aggregated	2
Day 3	Missing	Data checked and declared aggregated	2
Day 4	Missing	Data checked and declared aggregated	2
Day 5	32.5 mm	Data checked and declared aggregated	2
Day 6	0 mm	Data checked and declared good	0

11.6 Use case: Aggregation with instrument in maintenance

This refers to the case where, if we know that data missing over five days, with a known total on the sixth, includes a period where the station or sensor was undergoing maintenance[1], or observations were not possible due to some other cause, then we could distinguish in the daily messages between "data missing" during the maintenance period, where observations did not occur (say, on Days 2-3 in the example below), and aggregated (where we know that on days 4-5 measurements did occur, but we don't know the daily split). In the example provided below, assuming maintenance took place on Days 2-3, the sequence of flags would be: Day1 - aggregated; days 2-3 missing; days 4-5 aggregated (with the total aggregation on Day 5). The major distinguishing feature being that we know no data were available for two of the days in the sequence (Data missing – QC flag 5), and that the aggregation period represents the remaining periods flagged as aggregated (QC Flag 2).

In the example below, no value is available on Day 1, but measurement instruments are functional. On Days 2 and 3 the instruments are out of service, and therefore it is impossible to retrieve any data for these days. On Day 4, instrument is back in service, but no value is available. In Day 5, an aggregated value of 32.5mm is available, which includes the data from at least Days 4 and 5, and possibly also Day 1.

Coding:

Date	Precipitation	Associated QC information	QC code
Day 1	Missing	Data checked and declared aggregated	2
Day 2	Missing	Parameter is not measured at the station	5
Day 3	Missing	Parameter is not measured at the station	5
Day 4	Missing	Data checked and declared aggregated	2
Day 5	32.5	Data checked and declared aggregated	2

Note that this would obviously not include cases where data accumulated in Day 1, or any part of Days 2,3, were permanently lost. For instance, a raingauge washed away in a flood, or destroyed. In that case, the aggregation period would resume at Day 4, and the observations on Days 1, 2 and 3 would be flagged as 5.

[1] Note that Maintenance is intended as a generic term here. It can also include cases where observations were not possible due to some other causes, such as destruction of the instrument, an extended outage at an AWS, or some other cause.

Such a situation may also apply to other aggregated parameters, namely new snow depth, or Maximum/Minimum temperature. Again, following the example below, the aggregations for these parameters do not include any possible contributions from Days 2 and 3.

11.7 Use case: Aggregation with overflow/out of information range

Refers to cases where there is aggregated precipitation, and at the end, an amount of precipitation that exceeds the raingauge capacity is recorded. This may happen, for example, in cases where a heavy rainfall event occurs when the observer is absent (or needs to be evacuated!), or at isolated typically mountainous locations where a gauge is not checked every day, and orographic effects can lead to large accumulations.

Coding:

Date	Precipitation	Associated QC information	QC code
Day 1	3.5 mm	Data checked and declared good	0
Day 2	Missing	Data checked and declared aggregated	2
Day 3	Missing	Data checked and declared aggregated	2
Day 4	Missing	Data checked and declared aggregated	2
Day 5	203mm	Data checked and declared aggregated and out of instrument range	4

11.8 Use case: Solid precipitation aggregation with snowmelt over subsequent days

In this situation, heavy snow blocks an automated gauge, then subsequently melts over the next few days, producing spurious precipitation. In the following scenario, which happened at a high-altitude station in southeastern Australia in July 2022, heavy precipitation - mainly in the form of snow - was recorded over the first two days. The 38.8mm recorded on the first day was likely a minimum amount, and hence was allotted a flag of 3. Precipitation totals for Days 3 and 4 were known to have occurred on fine days with no precipitation, so the amounts on those days were purely snowmelt. On the fifth day there was a likely mix of snowmelt and new precipitation. Day 6 was believed to be entirely new precipitation.

Note that ideally, QC processes would cross-check against other data such as daily snow depth amounts and temperature information. However, if the snow blockage continues for a long period of time – say, in excess of one week – the likelihood is that even when the snow does melt, the total amount measured may be underestimated. In that case a QC flag of 4 (Aggregated and out of instrument range) may be appropriate.

Date	Precipitation	Associated QC information	QC code
Day 1	38.8 mm	Data checked and declared out of instrument range	3
Day 2	23.6 mm	Data checked and declared aggregated	2
Day 3	9.2 mm	Data checked and declared aggregated	2
Day 4	8.8 mm	Data checked and declared aggregated	2
Day 5	3.5 mm	Data checked and declared aggregated	2

Day 6	8.0 mm	Data checked and declared good	0

Note:
in this situation, it is suggested that cross-checking with other variables, notably "New snow", and temperature data, would make it clear that Days 3 and 4 in fact experienced no (new) precipitation, but that it would be strongly recommended that the NMHS document this situation as part of their national metadata.

11.9 Use case: missing complete set of values for daily computation

This case described a situation where not all values are available to compute “perfectly” a daily value, but the number of missing values is not in itself sufficient to trigger a missing value. Note that this does not apply only to mean temperature – it may be that, for instance, data are missing around the time of maximum or minimum temperature. In this case, coding should be based on the NMHS’s normal practices, utilizing also professional discretion as to the likely impact of the missing data. For instance:

- Where the calculation method to determine a mean daily temperature is to use all 1440 0ne-Minute values (60 min*24) from an AWS, but some are missing;
- Where the calculation method to determine a daily value (e.g. mean temperature) is to use 24 one-hourly values, but one or several are missing; or
- Where the calculation method to determine a daily value is to use the 8 main synoptic hours of the day (or four, or three), one or several values are missing; or
- An AWS is offline for a period where the minimum or maximum temperature might feasibly have occurred.

In these cases, the NMHS (data provider) may choose from several possibilities:

1. Notwithstanding the missing values, the NMHS data provider considers that the calculated value is still reliable. In that case the encoding would be:

Example:

Maximum Temperature	**Associated QC information**	**QC code**
10.4	Data checked and declared good	0

2. The data provider considers that the missing data may render the value suspect (e.g., night/day imbalance in observations used to compute mean temperature, or measurements missing around likely time of daily minimum/maximum temperature) -but that the value may still be sufficiently useful to share worldwide. In that case, the encoding would be:

Minimum Temperature	**Associated QC information**	**QC code**
10.4	Data checked and declared suspect	1

3. The data provider considers that the data is highly suspect, e.g., sufficient values are missing around the likely time of the minimum and maximum temperature that it is highly likely the true Maximum Temperature or Minimum Temperature is missed. In this case, it would be assumed that the available data does not represent a true assessment of the daily conditions, and therefore the data would be regarded as “doubtful”, i.e., to share it would be misleading. If this assessment is made by the Member’s NMHS, the coding would be:

Maximum Temperature	**Associated QC information**	**QC code**
Missing	Daily value not provided	6

12 Minimum requirements for international exchange

DAYCLI defines a set of **minimum requirements** to establish a baseline level of interoperability for international exchange. It also **supports progressive implementation by NMHSs**.

For a DAYCLI message to be exchanged internationally, the following elements are required:

- a **valid WIGOS Station Identifier** associated with the reporting station;
- a **clear definition of the Attribution day**, referenced to UTC;
- a **clear definition of the** measurement period or **Climatological day for each DAYCLI variable**;
- **at least one daily climate variable** reported according to national climatological practices.

Daily climate values should be provided for one or more parameters, depending on national data availability and validation status. The absence of some variables for a given station or day shall not prevent the exchange of a DAYCLI message.

Quality control remains a national responsibility. While the inclusion of **QC flags associated with daily values is strongly encouraged**, their absence shall not prevent international exchange. The lack of QC flags may reflect differences in national practices, limitations in mapping local QC information to DAYCLI representations, or a progressive implementation approach.

Similarly, the absence of optional metadata elements shall not prevent international exchange. This includes siting classification, measurement quality classification and the method to compute the mean daily temperature

NMHSs are encouraged to enhance the completeness of exchanged information over time, in line with national priorities and capacities.

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13 Transition from existing practices to DAYCLI

DAYCLI is designed to **complement existing national practices** for the production and dissemination of daily climate data, rather than to replace them. It supports a gradual transition towards a harmonized international exchange framework while preserving national responsibility for climatological definitions and processes.

In many NMHSs, daily climate data are produced from a combination of **manual observations and Automatic Weather Stations (AWS)** and managed within a Climatological Data Management System (CDMS). DAYCLI supports this diversity of observing systems and does not require changes to national observation or data processing workflows.

For NMHSs operating AWS networks, daily climate values are typically computed within the CDMS based on sub-daily observations received from AWS. The CDMS is responsible for applying national climatological rules, quality control procedures and validation steps before daily values are archived and prepared for exchange using DAYCLI.

NMHSs may implement DAYCLI progressively, starting with a limited number of stations, variables or metadata elements. This phased approach allows services to gain operational experience, refine workflows and expand coverage over time without disrupting existing national operations.

DAYCLI also supports continuity of long-term climate series during transitions in observing systems, station upgrades or changes in processing environments. By exchanging nationally validated daily values together with their context, DAYCLI helps preserve the integrity and interpretability of climate records across such transitions.

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14 From national datasets to DAYCLI BUFR implementation

DAYCLI is an **exchange format** derived from nationally validated daily climate datasets produced and managed under the responsibility of NMHS climatology departments. The implementation of DAYCLI does not modify national climatological values, definitions or quality assessments.

In a typical workflow, daily climate values are first measured or computed, quality controlled and validated by experts within a **Climatological Data Management System (CDMS)**, in accordance with national climatological practices. Once validated, these daily values constitute the authoritative national dataset from which DAYCLI messages are generated.

The transformation of national datasets into DAYCLI BUFR messages is a **technical encoding step** that supports international exchange. This step may be implemented within the CDMS itself or through external tools and software, provided that the content of the DAYCLI message faithfully reflects the nationally validated data and associated metadata.

DAYCLI BUFR encoding shall preserve the Attribution day definition, associated time periods, metadata and quality information established at national level. The encoding process shall not introduce reinterpretation or reprocessing of climatological values.

NMHSs may use existing BUFR libraries and tools, existing software or develop their own software solutions, to implement DAYCLI encoding and decoding. Examples of some software packages that enable this are provided in Section 17.2 below. By way of an example, Annex C below provides an illustrative example of how DAYCLI BUFR messages may be produced using existing software solutions. The choice of implementation approach remains a national decision and does not affect the interpretation of the exchanged data.

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15 Collection and use of DAYCLI data by regional and global centres

DAYCLI messages may be **collected, stored and made available** by regional and global climate data centres in support of climate monitoring, assessment and services. Such centres may include WMO Regional Climate Centres, Global Data Centres or other designated entities.

The collection and handling of DAYCLI data by regional or global centres does not alter the **national responsibility** of NMHSs for the content, validation and interpretation of the data they provide. DAYCLI represents the exchange of nationally validated daily climate values together with their associated metadata and quality information.

Regional and global centres may organise DAYCLI data within databases or information systems to facilitate access, dissemination and analysis at regional and global levels. This organisational processing shall not modify the original daily climate values provided by NMHSs.

Regional and global centres may perform additional analyses or quality assessments for specific applications, such as regional climate monitoring or global climate assessments. Any such processing shall be clearly documented and distinguished from the original nationally provided DAYCLI data.

Users of DAYCLI data disseminated by regional or global centers are encouraged to consider national metadata, quality information and Attribution day definitions. This is particularly important when comparing or combining data from different countries or regions. Differences in national practices, particularly with respect to Attribution day and time references, may influence interpretation.

DAYCLI supports the transparent use of daily climate data at regional and global levels while preserving the integrity, traceability and authority of national climate records.

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16 Annex A- Glossary

Attribution day
The **Attribution day** is the calendar day to which an NMHS assigns daily climate values, defined according to national climatological practices and referenced to a Local Meteorological Time Zone (LMTZ). An Attribution day is defined **for a given observing station and element**, recognizing that a national territory may include more than one LMTZ. The Attribution day provides the temporal reference used for daily climate data exchange and archiving.

BUFR
BUFR, Binary Universal Form for the Representation of meteorological data, is the name of a binary code for the exchange and storage of data.

Climate record
A set of long-term observations with time-series components covering any or all of the atmosphere, ocean, land and land water, cryosphere and space weather domains, with associated metadata, recorded according to standards and archived for climate purposes. It includes observations recorded on paper, as well as scanned and digitized records. The metadata document, at the minimum, the what, who, when, where and how an observation was made.

Climate Data Management System (CDMS)
A system used by NMHSs for the collection, quality control, validation, archiving and management of climate records. The CDMS constitutes the authoritative environment for the production of nationally validated daily climate datasets.

Climatological day
The 24 hour period (for instance midnight to midnight, 6am to 6am, 8pm to 8pm etc.; local time) over which a daily parameter has been accumulated or measured. Examples include the minimum, maximum and mean temperature, precipitation accumulation etc. For a given station, the Climatological day may vary with parameter.

Daily climate value
A value representing a climatological variable (e.g. temperature, precipitation, snow depth) associated with an Attribution day, derived from observations processed according to national climatological practices.

Daily mean temperature
A daily climate value representing the mean air temperature associated with an Attribution day, calculated according to national methods. Information on the calculation method must be provided as metadata where available.

Local Meteorological Time Zone (LMTZ)
The local time reference used by an NMHS to define climatological Attribution day and associated measurement time periods for daily climate data production. A national territory may include more than one LMTZ.

Measurement quality classification
A classification describing the overall quality of a measurement process and instrumentation, as defined in *WMO-No. 8*. It comprises four classes (A to D), where Class A represents the highest measurement quality.

National responsibility
The principle that NMHSs retain full authority over the definition, validation, interpretation and dissemination of their country's national climatological data.

OSCAR/Surface database
Is the **WMO's official global repository** for metadata on surface-based observing stations and platforms, including their location, instrumentation, and operational details. It supports the registration and management of **WIGOS Station Identifiers (WSI)** and ensures standardized international data exchange.

QC flag
A code used to describe the outcome of quality control applied to a daily climate value. It is included as part of the metadata, and provides information on the confidence one may place in the data, as well as other pertinent

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information about whether a stated daily value is aggregated, represents an overflow, etc. QC flags in DAYCLI represent the result of national quality control processes and do not replace national assessments.

Quality control (QC)
The set of procedures applied to climate data to detect, assess and document potential errors, inconsistencies or limitations, using automated checks and expert judgement in accordance with national practices.

Siting classification
A classification describing the exposure and environment of an observing station, as defined in the *Guide to Instruments and Methods of Observation (WMO-No. 8)*. It comprises five classes (Class 1 to Class 5), where Class 1 represents optimal siting conditions.

WIGOS Station Identifier (WSI)
is a unique identifier used to register an observing station or platform in the OSCAR/Surface database. It replaces the legacy WMO identifiers (5-digit numbers for synoptic and climate stations), which had limited capacity.

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17 Annex B- References

This annex lists the principal WMO regulatory and technical documents relevant to the implementation and use of the DAYCLI BUFR message. The references provide authoritative guidance on coding standards, observing practices, climatological data management and quality control.

17.1 WMO regulatory and technical documents

Manual on Codes (WMO-No. 306, Volume I.2, Edition 2026)

Guide to Instruments and Methods of Observation (WMO-No. 8, Volume I, Edition 2024)

Manual on the WMO Integrated Global Observing System (WMO-No. 1160, Edition 2024)

Guide to the WMO Integrated Global Observing System (WMO-No. 1165, Edition 2024).

Guide to Climatological Practices (WMO-No. 100, Fourth Edition, 2023)

Climate Data Management System Specifications (WMO-No. 1131, Edition 2025)

WMO Guidelines on Surface Station Climate Data Quality Control and Quality Assurance for Climate Applications (WMO-No. 1269, Edition 2021)

17.2 BUFR software and implementation resources

Some NMHSs and organisations are freely sharing their software, libraries and tools to support the encoding and decoding of DAYCLI BUFR messages. Some of these are listed below. This list is not exhaustive and does not imply endorsement of any particular solution.

Environment and Climate Change Canada
BUFR encoding and decoding library supporting climatological applications: https://github.com/ECCC-MSC/libecbufr

European Centre for Medium-Range Weather Forecasts (ECMWF)
Reference software library for encoding and decoding BUFR and GRIB messages, supporting C, Fortran and Python interfaces.
GitHub ecCodes: https://github.com/ecmwf/eccodes
Documentation and download: https://confluence.ecmwf.int/display/ECC/ecCodes+Home

National Institute for Space Research in Brazil (INPE)
Example implementation supporting DAYCLI BUFR encoding and decoding, provided as an open reference for NMHSs.

National Oceanic and Atmospheric Administration (NOAA)
BUFR processing library used operationally within NOAA environments: https://github.com/NOAA-EMC/NCEPLIBS-bufr

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18 Annex C-Example implementation of DAYCLI production using existing software (INPE in Brazil)

This annex provides an illustrative example of how DAYCLI BUFR messages may be produced using existing software solutions. The example is intended to support understanding and experimentation and does not imply endorsement of any specific implementation, software package or operational approach.

Further information on this software is available at: https://github.com/sergioh-pessoal/daycli_toolkit (or via the National Institute for Space Research in Brazil (INPE) distribution link).

18.1 Context and objectives

The example implementation is based on software developed by the Brazilian National Institute for Space Research (INPE) to support the encoding and decoding of DAYCLI BUFR messages. The software is designed to operate within typical NMHS climatological workflows and to interface with data produced and validated by a Climatological Data Management System (CDMS).

The objective of this example is to demonstrate how nationally validated daily climate data can be transformed into DAYCLI BUFR messages without modifying national climatological practices, definitions or quality control procedures.

18.2 Position within NMHS workflows

In this example, daily climate values are computed, quality controlled and validated within the CDMS in accordance with national climatological practices in each NMHS. Once validated, the daily values constitute the authoritative national dataset.

The DAYCLI software operates as a technical encoding layer**,** converting nationally validated daily climate data and associated metadata into DAYCLI BUFR messages for international exchange. The software does not perform climatological calculations or quality control decisions.

18.3 Input data formats and handling of observation time practices

The example implementation provides flexibility in the preparation of input files supplied by NMHSs by supporting **two alternative input data formats**, reflecting different observing time practices encountered in national and historical datasets.

The first input format is intended for stations whose observations are performed at **fixed and consistent times throughout the year**, in line with standard WMO observing practices. This format is well suited to operational surface stations where observation schedules do not vary seasonally.

The second input format is designed to accommodate stations for which **observation times may vary during the year**, for example due to changes between summer and winter time. While such practices are **not recommended by WMO** for climatological observations, they may be encountered in historical datasets or in data originating from stations outside the WMO observing network that nevertheless provide valuable climate information.

By supporting both input formats, the example implementation facilitates the integration and preservation of diverse national and historical datasets, while allowing the resulting DAYCLI BUFR messages to clearly document Attribution day and associated measurement time periods. This flexibility supports data continuity and transparency without altering national climatological values or practices.

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18.4 Examples of input files for DAYCLI BUFR production

The input data formats are based on a **Fortran namelist structure**, in which groups of information are defined using blocks starting with a group name (e.g. `$GroupName`) and ending with a slash (`/`).

The initial groups contain fixed metadata, such as:

- generating centre and sub-centre identifiers;
- station location information;
- fixed measurement periods associated with each parameter.

The final group, `$DATA_SECTION`, contains the actual daily climate values and associated quality flags, separated by semicolons (`;`).

18.4.1 Format 1 – Fixed observation times

In Format 1, the measurement periods are defined in separate groups and are not included in the data section. For example, for maximum temperature:

```
&STIME_TX

HOUR=06

MINUTE=01

DT=0

/
```

In this example:

`HOUR` and `MINUTE` define the observation time;

`DT = 0` indicates that the measurement period starts on the current Attribution day (LMTZ);

`DT = -1` indicates that the measurement period starts on the previous day (LMTZ).

The data section then contains only the reference date and the sequence of values and quality flags:

```
&DATA_SECTION

!date ;rr(mm); qrr;  ds;qds;tsd;qtsd;tn(c);qtn; tx(c);  qtx;tm(c);qtm

20220101 ;2.9;   0;   ; 5 ;   ; 5 ;   23.0; 0;   31.9;    0;25.9; 0

20220102 ;7,4;   0;   ; 5 ;   ; 5 ;   23,4; 0;   32.3;    0;26.3; 0

20220103 ;6.0;   0;   ; 5 ;   ; 5 ;   22.8; 0;   28.4;    0;24.7; 0

20220104 ;3.3;   0;   ; 5 ;   ; 5 ;   21.7; 0;   32.1;    0;25.8; 0

20220105 ;1.4;   0;   ; 5 ;   ; 5 ;   22.8; 0;   31.5;    0;26.4; 0

20220106 ;0.2;   0;   ; 5 ;   ; 5 ;   22.9; 0;   31.9;    0;26.3; 0

20220107 ;0.1;   0;   ; 5 ;   ; 5 ;   23.3; 0;   31.7;    0;26.9; 0

20220108 ;7.4;   0;   ; 5 ;   ; 5 ;   22.5; 0;   31.1;    0;25.9; 0

/
```

Where:

- `rr` = total accumulated precipitation
- `ds` = fresh snow
- `tsd` = total snow depth
- `tn` = minimum temperature
- `tx` = maximum temperature
- `tm` = mean temperature

Quality flags are indicated using the prefix `q` (e.g. `qrr`, `qtn`, etc.).

The software uses the fixed observation times and the reference date to compute the start and end times required for DAYCLI BUFR encoding.

18.4.2 Format 2-Explicit measurement intervals

In Format 2, the measurement periods are explicitly defined within the data section. Each parameter is associated with its own time interval, value and quality flag.

```
&DATA_SECTION
!!date;interval;tsd;qtsd;interval;rr;qrr;interval;fsd;qfsd;interval;tx;qtx;interval;tn;qtn;interval;tm;qtm

20220401;20220401T0601;0;7;20220401T0601-20220402T0600;15.5;0;20220401T0601-
20220402T0600;0;7;20220401T0601-20220402T0600;278.95;0;20200331T1801-
20220401T1800;273.35;0;20220401T0001-20220402T0000;275.05;0

20220402;20220402T0601;0;7;20220402T0601-20220403T0600;1.9;0;20220402T0601-
20220403T0600;1;7;20220402T0601-20220403T0600;277.65;0;20220401T1801-
20220402T1800;274.05;0;20220402T0001-20220403T0000;275.85;0

/
```

In this case, the software directly encodes the provided time intervals into the DAYCLI BUFR message without performing additional calculations.

18.5 Additional information

Further details and updates are available via the software repository.

19 Annex D Siting and measurement quality classification for temperature

0	Reserved
1	1A
2	1B
3	1C
4	1D
5	Reserved
6	2A
7	2B
8	2C
9	2D
10	Reserved
11	3A
12	3B
13	3C
14	3D
15	Reserved
16	4A
17	4B
18	4C
19	4D
20	Reserved
21	5A
22	5B
23	5C
24	5D
25	Reserved
26	1
27	2
28	3
29	4
30	5
31	A
32	B
33	C
34	D
35–254	Reserved

Notes:

(1) Code figures 1–24 are siting classifications according to the *Guide to Instruments and Methods of Observation* (WMO-No. 8), Volume I, Chapter 1, Annex 1.D and Annex 1.G.

(2) Code figures 26–30 are siting classifications according to the *Guide to Instruments and Methods of Observation* (WMO-No. 8), Volume I, Chapter 1, Annex 1.D.

(3) Code figures 31–34 are measurement quality classifications according to the *Guide to Instruments and Methods of Observation* (WMO-No. 8), Volume I, Chapter 1, Annex 1.G.

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20 Annex E Siting and measurement quality classification for precipitation

Code table 0 08 096

Code figure	
0	Reserved
1	1A
2	1B
3	1C
4	1D
5	Reserved
6	2A
7	2B
8	2C
9	2D
10	Reserved
11	3A
12	3B
13	3C
14	3D
15	Reserved
16	4A
17	4B
18	4C
19	4D
20	Reserved
21	5A
22	5B
23	5C
24	5D
25	Reserved
26	1
27	2
28	3
29	4
30	5
31	A
32	B
33	C
34	D
35–254	Reserved

Notes:

(4) Code figures 1–24 are siting classifications and measurement quality classifications according to the *Guide to Instruments and Methods of Observation* (WMO-No. 8), Volume I, Chapter 1, Annex 1.D and Annex 1.G.

(5) Code figures 26–30 are siting classifications according to the *Guide to Instruments and Methods of Observation* (WMO-No. 8), Volume I, Chapter 1, Annex 1.D.

(6) Code figures 31–34 are measurement quality classifications according to the *Guide to Instruments and Methods of Observation* (WMO-No. 8), Volume I, Chapter 1, Annex 1.G.

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